gpeyt1-08 · CD triaxial strength

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A consolidated-drained (CD) triaxial test on a remolded clean sand ($c' = 0$) consolidated to $\sigma'_3 = 20$ psi fails at a deviator stress of 52 psi. Pore pressure was zero throughout; specimen diameter 2.8 in., height 5.6 in.; axial strain at failure 6%. A second specimen of the same sand will be tested at $\sigma'_3 = 35$ psi.

Most nearly, the deviator stress at failure expected for the 35 psi test is:

A. 91 psi
B. 126 psi
C. 99 psi
D. 45 psi

Given equations / data

> $\sin\varphi' = (\sigma'_1 - \sigma'_3)/(\sigma'_1 + \sigma'_3)$; $\sigma'_1 = \sigma'_3 \tan^2(45 + \varphi'/2)$ for $c' = 0$

Worked solution

1. Test 1: $\sigma'_1 = 20 + 52 = 72$ psi
2. $\sin\varphi' = 52/92 = 0.5652 \rightarrow \varphi' = 34.4°$
3. Kp-type ratio $\tan^2(45 + \varphi'/2) = 3.600$
4. Test 2: $\sigma'_1 = 35 \times 3.600 = 126.0$ psi; $\Delta\sigma_f = 126.0 - 35 = 91.0$ psi

Final answer

91 psi

Why the other values are wrong

- **126 psi** — reported σ'_1 instead of deviator stress
- **99 psi** — used $\tan\varphi = \Delta\sigma/\sigma_1$
- **45 psi** — reported the Mohr-circle radius (½ deviator)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3.2 Shear Strength — Effective Stress (Mohr–Coulomb), p. 87

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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